

Lecture 2 Module 1

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Functions. A, B

$$f: A \rightarrow B$$

$$f \subseteq A \times B \quad (x, y) \in f := f(x) = y$$

• for each $x \in A$, there is a $y \in B$

→ s.t. $f(x) = y$: existence of 'image'

• for each $x \in A, y, y' \in B$

$$f(x) = y, f(x) = y' \Rightarrow y = y'$$

: uniqueness of image.

• f is one-one / injective
for each $x, x' \in A, f(x) = f(x')$
 $\Rightarrow x = x'$

$$\text{or } x \neq x' \text{ in } A \Rightarrow f(x) \neq f(x')$$

• f is onto / surjective:

for each $y \in B$, there is $x \in A$
s.t. $f(x) = y$.

• Bijection $A \sim B$

Obs. $f(A) := \{y \in B : f(x) = y, \text{ } \neg$

for some $n \in A$

• f is onto iff $f(A) = B$.

• $A \sim f(A)$; if f is one-one.

• Composition of functions:

$$f: A \rightarrow B, g: B \rightarrow C$$

$$g \circ f: A \rightarrow C$$

$$g \circ f(a) := g(f(a)), \text{ for each } a \in A.$$

• ' $\circ$ ': associative, not commutative
: preserves one-one, onto, bijection

• $f^{-1}: \mathcal{P}(B) \rightarrow \mathcal{P}(A)$ s.t. for any

$$Y \subseteq B, f^{-1}(Y) = \{a \in A: f(a) \in Y\}$$

– inverse image of Y under f

Ex. 1. f is onto iff $f^{-1}(Y) \neq \emptyset$ for each
 $Y (\neq \emptyset) \subseteq B$.

2. f is one-one iff

$f^{-1}(\{y\})$ is a singleton, for each
 $y \in B \cap f(A)$.

• For one-one, onto functions $f: A \rightarrow B$,

$$f^{-1}: B \rightarrow A$$

$$f^{-1}(b) = a \text{ iff } f(a) = b.$$

Other characterisations: ~